MILITARY SPECIFICATION

BOOSTER, M125A1 ASSEMBLY LESS BOOSTER PELLET AND LEAD CUP CHARGE

1. SCOPE

1.1 This specification covers one type of booster for use with high explosive ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F8595541	- Booster, M125A1 Assembly
C7548175	- Box, Packing Ammunition, for Booster Assembly, Details and Packing

FSC 1390

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2.1 (Cont'd)

C7548174 - Carton Packing
IEL-F8595541 - Inspection Equipment List for Booster, M125A1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on applicable drawings, referenced standards, and specifications except as provided in 6.1.

3.2 Environmental.

3.2.1 The booster shall comply with MIL-STD-331, Jolt, test No. 101.

3.2.2 The booster shall comply with MIL-STD-331, Jumble, test No. 102.

3.2.3 The booster shall comply with MIL-STD-331, Transportation - Vibration, test No. 104.

3.3 Ballistics (See 6.3.2)

3.3.1 The booster shall not function prematurely. (See 6.3.1)

3.3.2 Target impact nonfunction distance.-The booster shall not detonate the round upon impact element function when fired in the 90MM Gun, Service Charge, (2400 feet per second, f.p.s.) on a target 200 plus 5 feet from the muzzle of the weapon.

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3.3.3 Target impact function distance.-The booster shall detonate the round upon impact when fired in the 105MM Howitzer, Charge 7 (1550 f.p.s.) on a target 400 minus 5 feet from the muzzle of the weapon.

3.3.4 Ground or water impact.-The booster shall detonate the round upon impact when fired in the 155MM Howitzer, Charge 1 (680 f.p.s.) at an impact angle of 45° minimum.

3.4 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.4.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawings.

3.4.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.4.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.4.5 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The method and agents used in processing cleaning shall not adversely affect the finish or contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.-When required by the contract, the first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

40 component parts as delineated in 4.2.2 (aggregate)
200 subassemblies as delineated in 4.2.3 (aggregate)
73 assemblies as delineated in 4.2.5 and 4.2.6 (aggregate)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, sixty (60) booster assemblies shall be forwarded to a designated proving ground as specified in the contract. Forty (40) component parts, 200 subassemblies and 13 booster assemblies shall be inspected as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each component shall be tested as in 4.3.3.1 (Hardness).

4.2.3 Subassemblies.

4.2.3.1 Fifty (50) of each subassembly shall be tested as in 4.3.3.2 (Static load).

4.2.4 Acceptance criteria.-Failure of any component or subassembly to comply with the requirements of the applicable Drawing shall be cause for rejection of the initial production sample.

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4.2.5 Booster assemblies.-Thirteen (13) booster subassemblies shall be subjected to the following MIL-STD tests:

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.5.1	4.2.5.1
Jumble	4	3.2.2	4.4.5.2	4.2.5.1
Transportation-Vibration	5	3.2.3	4.4.5.3	4.2.5.1

4.2.5.1 Acceptance criteria.-The criteria of the applicable MIL-STD apply as determined by disassembly and examination in accordance with the procedure in 4.4.5. Where accept/reject decision is based on engineering judgement, concurrence by the responsible Technical Agency must be obtained.

4.2.6 Ballistics.-Sixty (60) boosters, of the initial production sample, shall be subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Impact Nonfunction Distance	20	3.3.2	4.4.6.1
Impact Function Distance	20	3.3.3	4.4.6.2
Ground or Water Impact	20	3.3.4	4.4.6.3

4.2.6.1 Acceptance criteria.-The occurrence of the following shall be cause for rejection of the initial production sample:

a. A premature burst in any phase. (See 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Phase.

c. Failure of more than one round total to function in the combined following phases:

(1) Target Impact Function Phase
(2) Ground or Water Impact Phase.

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4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

4.3.1.1 Lot formation.-Each inspection lot of boosters shall be comprised of components (except explosive components) each produced by one manufacturer, by one process, in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled boosters shall be limited to one manufacturer and one interfix number.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a single defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing examination for visual defects.

4.3.2.1 Classification of defects.-Inspection of boosters shall be in accordance with the following:

4.3.2.2 Cup, Booster, Drawing B8595509 covering a detail of Drawing F8595541.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR	AQL = 1.0%	
101.	Inside depth, improper	Gage
102.	Pitch diameter of thread, improper	Gage
103.	Minor diameter of thread exceeds max.	Gage
104.	Outside diameter, improper	Gage
105.	Effective length of thread, less than min.	Gage
106.	Metal defective	Visual
107.	Cut, score, or gouge on inside wall	Visual
108.	Foreign matter	Visual

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4.3.2.3 Cup, Closing, Drawing B8595508, covering a detail of Drawing F8595541.

Defect	Method of Inspection
CRITICAL	
1. Split, cut or crack	Visual
MAJOR AQL = 1.0%	
101. Length under flange, improper	Gage
102. Outside diameter exceeds max.	Gage
103. Thickness of base	Gage
104. Foreign matter	Visual

4.3.2.4 Body, Drawing F8595511 covering a detail of Drawing F8595541.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Overall length of body exceeds max.	Gage
102. Diameter of M hole below min.	Gage
103. Staking hole in thread, missing	Visual
104. Burr	Visual
105. Metal defective	Visual

4.3.2.5 Balance Staff, Escape Wheel, No. 1 and No. 2 Gear and Pinion Assemblies Drawing D8594614, C8595538, C8594673 and C8594681 covering a detail of Drawing F8595541.

CRITICAL

None defined

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Eccentricity of gears and escape wheel, improper	Gage
102. Perpendicularity of gears and escape wheel, improper	Gage
103. Escape wheel contour, improper	Gage
104. Pitch diameter of gears, improper	Gage
105. Broken or missing teeth, deformed or bent assembly	Visual
106. Gears, balance staff or escape wheel inadequately staked or seated	Visual

4.3.2.6 Booster, subassembly, Drawing F8594668 covering a detail of Drawing F8595541.

CRITICAL	
1. Interference condition which precludes return of either spring-loaded detent from the fully open position to the position of engagement in the rotor recess. (Check prior to nonarming test)	Visual/Manual
2. Rotor not locked in unarmed position (See 6.2.3) (Check immediately prior to packing)	Visual/Manual
MAJOR AQL = .40%	
101. Pitch diameter of large external thread, improper	Gage
102. Major diameter of large external thread, improper	Gage
103. Pitch diameter of small external thread, improper	Gage
104. Major diameter of small external thread, improper	Gage
105. Pitch diameter of internal thread, improper	Gage
106. Minor diameter of internal thread, exceeds max.	Gage
107. Effective length of internal thread, less than min.	Gage
108. Length from base of body to top of plates including all protrusions, exceeds max.	Gage
109. Loose cover plate	Visual/Manual
110. Any part missing	Visual
111. Any part broken or deformed	Visual
112. Staking hole missing or inadequate	Visual
113. Damaged threads	Visual
114. Burr	Visual
115. Foreign matter	Visual

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4.3.2.7 Booster, Assembly F8595541 (less lead charge and booster pellet).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Nomenclature missing, incorrect or illegible	Visual

4.3.2.8 Carton, packing, Drawing C7548174.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Assembly damaged so that contents are exposed or liable to become exposed	Visual
102. Sealing strip incomplete, badly wrinkled, or not in contact with surface of containers for at least $\frac{1}{4}$ inch around entire circumference	Visual
MINOR AQL = 2.5%	
201. Contents move when shaken	Manual
202. Marking misleading, incorrect or illegible	Visual
203. Label missing	Visual

4.3.2.9 Box, packing, Drawing C7548175.

CRITICAL	
None defined	

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Board broken	Visual
102. Hardware missing, broken or loose	Visual/Manual
103. Contents exposed	Visual
104. FSC symbol misleading, incorrect or illegible	Visual
105. Strapping missing, broken or loose	Visual/Manual
MINOR AQL = 2.5%	
201. Hardware improperly applied or engaged	Visual/Manual
202. Contents move when shaken	Manual
203. Car seal missing, unsealed or improperly positioned	Visual/Manual
204. Marking misleading, incorrect or illegible	Visual/Manual

4.3.3 Inspection testing.-The following test shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (Destructive).-Twenty (20) samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot (See 4.4.1).

Component	Drawing Number
Wheel, Escape	C8595539
* Pin, Balance	C8595540

* In lieu of tensile strength certification

4.3.3.2 Static load (Destructive).-Fifty (50) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.2).

Component	Drawing Number
No. 1 Gear and Pinion Assembly	C8594673
No. 2 Gear and Pinion Assembly	C8574681
Escape Wheel Assembly	C8595538
Balance Staff Assembly and Pins	C8594614

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4.3.3.3 Arming - (Nondestructive).-Each booster assembly shall be subjected to the arming test. Failure of the rotor to become locked-in with the detonator in alignment with the flash hole, when spun at the Arming Spin Requirement on Drawing F8594688, shall be cause for rejection of the assembly (See 4.4.3).

4.3.3.4 Non-Arming (Nondestructive).-Each booster assembly shall be subjected to a non-arming test. Failure of either one or both spring loaded detents to remain locked in the rotor, when subjected to the Non-Arm Spin Requirement of Drawing F8594688, shall be cause for rejection of the assembly (See 4.4.4).

4.3.3.5 Environmental tests (Destructive).-Thirteen (13) booster sub-assemblies from each production lot shall be subjected to the following MIL-STD-tests (See 4.4.5).

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.5.1	4.3.3.5.1
Jumble	4	3.2.2	4.4.5.2	4.3.3.5.1
Transportation-Vibration	5	3.2.3	4.4.5.3	4.3.3.5.1

4.3.3.5.1 Acceptance criteria.-The criteria of the applicable MIL-STD apply as determined by disassembly and examination in accordance with procedure in 4.4.5. Where accept/reject decision is based on engineering judgement, concurrence by the responsible Technical Agency must be obtained.

4.3.3.6 Ballistics (Destructive).-Sixty (60) boosters from each lot shall be subjected to the following ballistic test phases: (See 4.4.6).

Phases Ambient	Quantity	Required	Test Method
Impact Nonfunction Distance	20	3.3.2	4.4.6.1
Impact Function Distance	20	3.3.3	4.4.6.2
Ground or Water Impact	20	3.3.4	4.4.6.3

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4.3.3.6.1 Acceptance criteria.-The occurrence of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature burst in any phase (See 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Phase.

c. Failure of more than one round total to function in the combined following phases:

(1) Target Impact Function Phase.
(2) Ground or Water Impact Phase.

4.3.4 Inspection equipment.-The examinations and tests shall be made using equipment listed on IEL-8595541. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test method and procedures.

4.4.1 Hardness.-The test shall be performed using the value as specified on the applicable drawing. The sample units shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual reading. All tested samples shall be scrapped.

4.4.2 Static load.-The sample units shall be tested to the value specified on the applicable drawing. All units subjected to the static load test shall be scrapped.

4.4.3 Arming.-The unarmed booster assembly shall be spun for a period of 10 seconds maximum in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on Drawing F8594668. After the test has been performed, the rotor shall be returned to the unarmed position (See 6.2.1 and 6.2.3).

4.4.4 Non-Arming.-The unarmed booster (See 6.2.3) shall be spun in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on Drawing F8594668. The non-arming test shall be last spin test to which the booster is subjected prior to packing.

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4.4.5 Environmental test procedures.-The component parts of the booster, which have been disassembled and inspected following tests, shall be scrapped.

4.4.5.1 Jolt.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101. The booster shall be disassembled and inspected as follows:

a. Remove dust cover check position of rotor.

b. Subject the booster to the nonarming requirements of the applicable drawing.

c. Remove cover plate

d. Check for broken, deformed or displaced parts (that would render the booster unsafe to handle or dangerous to use).

4.4.5.2 Jumble.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102. The booster shall be disassembled and inspected as follows:

a. Conditions listed in 4.4.5.1

4.4.5.3 Transportation-Vibration.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104.

a. Remove dust cover check position of rotor.

b. Subject the booster to the non-arming and arming requirements of the applicable drawing.

c. Conditions listed 4.4.5.1 (c to d)

4.4.6 Ballistics.-The booster assembly, Drawing F8595541 shall be assembled to fuze, PD, 48A3 then assembled to a suitable projectile and fired as follows.

4.4.6.1 Impact nonfunction distance phase.-The test sample shall be fired as specified in 3.3.2 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.6 as appropriate.

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4.4.6.2 Impact function distance phase.-The test sample shall be fired as specified in 3.3.3 against a suitably supported one inch plywood screen of adequate size under condition delineated in 4.2.6 or 4.3.3.6 as appropriate.

4.4.6.3 Ground or water impact phase.-The test sample shall be fired as specified in 3.3.4 under the conditions as delineated in 4.2.6 or 4.3.3.6 as appropriate.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking.-Packaging, packing, labeling and marking shall be as specified on Drawing C7548175 with the following exception. Closing Cups, Drawing B8595508 and Booster Cups, Drawing B8595509 may be shipped separately in accordance with best commercial practices as an alternate procedure.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 and those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate booster performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on booster functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the booster.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

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6.2 Armed/Unarmed Boosters.

6.2.1 Armed.-Rotor is locked by the spring loaded rotor lock pin in the position in which the detonator is fully visible through the flash hole.

6.2.2 Partially armed.-Rotor not locked by both spring-loaded detents.

6.2.3 Unarmed.-Rotor locked by both spring loaded detents with the detonator not at all visible through the flash hole.

6.3 Ballistic parameters.

6.3.1 Premature burst.-Shall be defined as the occurrence of any of the following:

a. If the round detonates in the bore of the weapon

b. If the round detonates in flight within 200 feet from the muzzle of the weapon

6.3.2 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390 -A735

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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